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Honeywell Process Solutions

APM/HPM I/O Simulator
User's Guide

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R660

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About This Document

This APM/HPM I/O Simulator manual provides a guide to installing, operating, and using the Advanced Process Manager and or High-Performance Process Manager simulator software. This manual also describes how the simulator software is supported in the R500 and later versions of both the Advanced Process Manager and the High-Performance Process Manager.

Prior experience in configuring the APM or the HPM data points and using the APM or the HPM status displays is recommended for users of this product. For users requiring the APM or the HPM configuration information, see related publications provided in References.

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Symbol	Definition
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground: Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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Acronyms

Acronym	Expansion
AO	Analog Output
APM	Advanced Process Manager
APMM	Advanced Process Manager Module
DI	Digital Input
DO	Digital Output
EB	Exception Build
HLAI	High Level Analog Input
HM	History Module
HPM	High-Performance Process Manager
HPMM	High-Performance Process Manager Module
IDF	Intermediate Data Files
IOP	Input Output Processors
LCN	Local Control Network
LLAI	Low Level Analog Input
LLMUX	Low Level Analog Input Multiplexor
NAN	Not A Number
NIM	Network Interface Module
PI	Pulse Input
PWM	Pulse Width Modulated
SI	Serial Interface
SOE	Sequence of Events
STI	Smart Transmitter Interface

Acronyms

Acronym	Expansion
TAC	Technical Assistance Center
UCN	Universal Control Network
US	Universal Station

Parameters

Parameter	Expansion
OFFNORML	Off Normal Alarm
OPFINAL	Output Final
OPTDIR	Output Direct/Reverse Action
PV	Process Variable
PVHH	PV High High Alarm
PVHI	PV High Alarm
PVLL	PV Low Low Alarm
PVLO	PV Low Alarm
PVRAW	Raw PV (in EUs if Simulation Mode)
PVROCN	PV Rate of Change Negative
PVROCP	PV Rate of Change Positive
PVSOURCE	PV Source
TF	PV Filtering Lag Time

References

Publication title
Data Entity Builder Manual
Advanced Process Manager Implementation Guidelines
Process Manager/Advanced Process Manager Installation
Process Manager/Advanced Process Manager Checkout
Advanced Process Manager Specification and Technical Data
Advanced Process Manager Control Functions and Algorithms
Advanced Process Manager Parameter Reference Dictionary
Control Language/Advanced Process Manager Data Entry
Control Language/Advanced Process Manager Reference Manual
Advanced Process Manager Configuration Forms
Process Operations Manual
Process Manager/Advanced Process Manager Service
PM Family Parameter Reference Dictionary
High-Performance Process Manager Implementation Guidelines
High-Performance Process Manager Installation
High-Performance Process Manager Checkout
High-Performance Process Manager Specification and Technical Data
High-Performance Process Manager Control Functions and Algorithms
Control Language/High-Performance Process Manager Data Entry
Control Language/High-Performance Process Manager Reference Manual
High-Performance Process Manager Configuration Forms
High-Performance Process Manager Service

1. Read Me First

1.1 Requirements for Using the Simulator

Introduction

Before continuing in this manual, it is important that you understand the requirements of the Advanced Process Manager and High-Performance Process Manager I/O Simulator. It is also important that you complete the User registration form.

Registration responsibilities

Please take the following steps now to ensure support from Honeywell:

- Familiarize yourself with the enclosed Software License Agreement, Copyright Statements, Trademarks, and Credits.
- Complete the User Registration Form, sign it, and mail to the address shown.

Hardware requirement

The minimum hardware you need for the installation and operation of the APM/HPM I/O Simulators is as follows:

- Local Control Network (LCN)
- Network Interface Module (NIM)
- Universal Station (US)
- Advanced Process Manager (APM)
- High-Performance Process Manager (HPM)

A History Module (HM) is not required, but provides ease of use.

Software requirement

Release 500 and later versions of LCN and UCN software are required.

Compatibility

The APM/HPM I/O Simulators are compatible with LCN Release 500 and later.

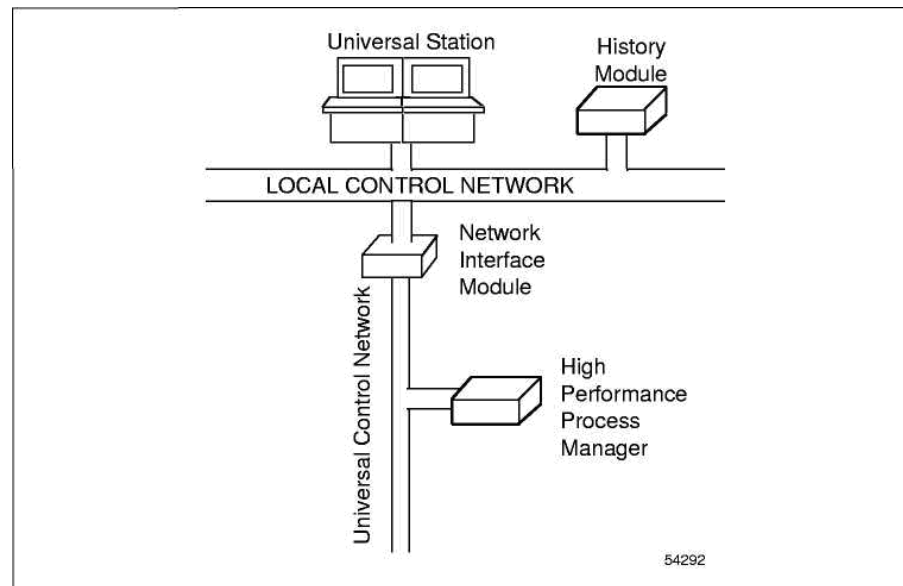
Example system

Figure 1 represents an example system configured to support the or the High-Performance Process Manager I/O Simulator.

1. Read Me First

1.1. Requirements for Using the Simulator

Figure 1 Example System Configuration



ATTENTION

It is not necessary to have both the APM and HPM nodes.

1.2 Software Registration

Registration form

Retain this copy of your software registration for your files.

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COMPANY

MAIL ADDRESS

(CITY) (STATE) (ZIP)

PLANT OR MILL SITE

(NUMBER) (STREET ADDRESS)

MAIL ADDRESS

(CITY) (STATE) (ZIP)

PHONE ()

AREA CODE

SERIAL NUMBER

DATE PURCHASED

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AUTHORIZED SIGNATURE

DATE

RETAIN THIS COPY IN YOUR FILES

1. Read Me First

1.2. Software Registration



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1. Read Me First

1.2. Software Registration

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2. Overview of the APM/HPM I/O Simulator

2.1 Overview of the I/O Functions

Intended uses

Both APM and HPM I/O Simulator personalities provide you the capability to simulate functions of the APM or HPM's Input/Output Processors (IOPs). They support a low-cost, high-fidelity approach to simulation intended for use in control strategy checkout or as a support tool in an operator-training environment.

Simulated I/O types

The I/O Simulation Personalities simulate the operation of all APM and HPM IOP modules, including:

- High Level Analog Input (HLAI)
- Smart Transmitter Interface (STI)
- Low Level Analog Input (LLAI)
- Low Level Analog Input Multiplexor (LLMUX)
- Analog Output (AO)
- Pulse Input (PI)
- Digital Input (DI)
- Digital Output (DO)
- Serial Interface (SI)
- High density IOPs (DO_32 and AO_16)



ATTENTION

The installed IOP and associated hardware are not required for operating the IOP simulation software.

Supported functions

The APM/HPM I/O simulation personality supports the following functions:

- Configuration of I/O database without requiring physical IOPs.

2. Overview of the APM/HPM I/O Simulator

2.1. Overview of the I/O Functions

- Real-time simulation of Primary IOP functions including data owner logic and alarm processing.
- Checkpoint save and restore of simulated Process Manager database—PVRAW (AI, DI Status/latched), OP (AO, DO), AV (DI Accumulation).
- Checkpoint restore of simulation database to a target Process Manager and I/O.
- Write access to input values (PV for all input types, PVRAW for AI, AV and PVFL for DI) for all input type modules.
- Pause and resume capability to interrupt and restart simulations.
- IOP hard failure simulation.
- Support for peer-to-peer communication.
- Support for physical SI IOP.

The communication processor handles all references to the I/O database internally. In a redundant configuration, the simulation personality must be installed on both the nodes.

Non-supported functions

The APM/HPM I/O simulator does not support the rebroadcast of alarms (R660 and later). For example, when an HPM is loaded with the simulator personality, the offnormal alarm disappears during a NIM fail over.

Simulation mode overview

When operating in the simulation mode, physical I/O is not required, and if present, is not used by either the APM or the HPM. Program access to read the simulated output data and to write the simulated PV data is enabled. This allows a user program in a DEC VAX (connected by way of a Computer Gateway), or an Application Module, or an APM or HPM to perform the process simulation function. Figure 2 and Figure 3 represent an overview of the APM and HPM simulator software.

2. Overview of the APM/HPM I/O Simulator

2.1. Overview of the I/O Functions

Figure 2 APM Simulator Overview

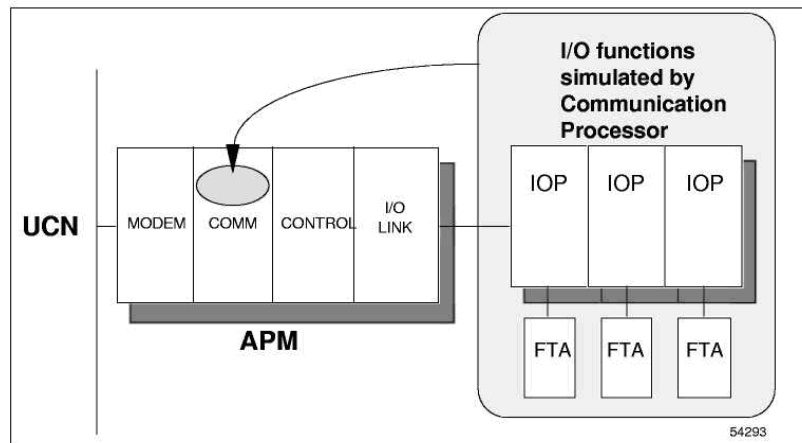
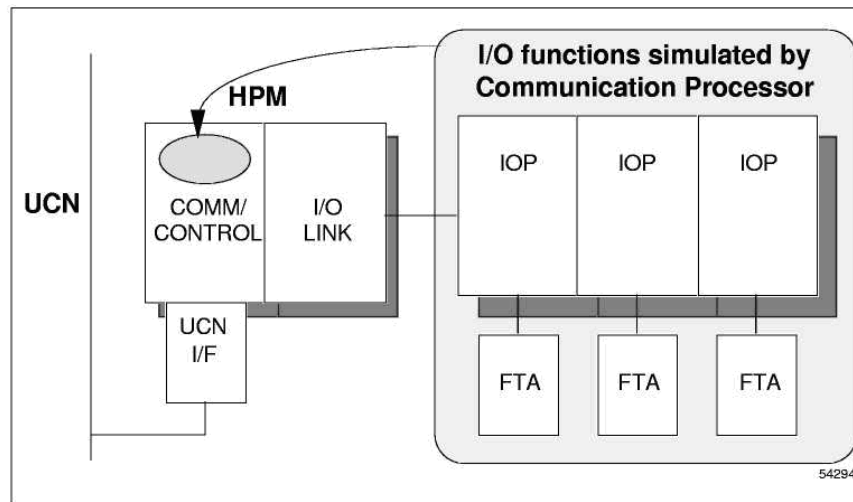


Figure 3 HPM Simulator Overview



2. Overview of the APM/HPM I/O Simulator

2.1. Overview of the I/O Functions

Definition of terms

You may wish to write your own simulation program to complement the functionality provided by the Process Manager I/O simulation personality. This document occasionally refers to a customer-written process-simulation program as "I/O simulation software," in order to distinguish it from Honeywell's I/O simulation personality option.

2.2 Simulated IOP Functions

Summary of functions

The simulation personality provides all IOP functions needed to enable your control strategy checkout. However, some time-related functions (such as filtering) use default values and are not supported when the APM or HPM is in a simulation personality. The database support (that is, configuration) is available for non-supported functions. A summary of key functions simulated, as well as functions not supported, is provided.

Analog input modules

The analog input modules (HLAI, STI, LLAI, LLMUX) have the following functions simulated:

- An operator or program is permitted to make stores to PVRAW. When an APM or an HPM is in the simulation mode, the PVRAW value is in engineering units. See Figure 4 Overview of Analog Input Simulator.
- The PVRAW value may be set to NAN (that is, Not A Number, —) to simulate a sensor or transmitter input failure.
- The alarms are detected and processed for the following IOP alarms—PVHI, PVLO, PVHH, PVLL, and BADPV. This includes alarm priority, enabling/disabling, and contact cutout functions. (PVROCP or PVROCN are not supported, although database configuration is supported.)
- The PV range checking and the PVSOURCE selection functions are supported.

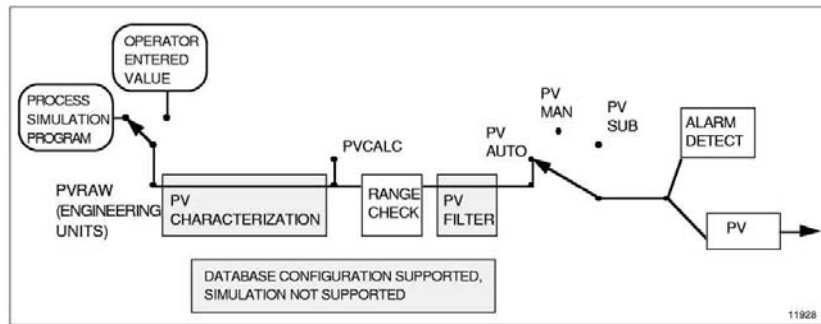
The following functions are not simulated:

- PV Characterization: As a user-written process simulation program (that is, I/O stimulation software) normally stores to PVRAW in engineering units, the PV characterization is not needed. When the PVRAW value is stored, the value is transferred directly to PVCALC.
- PV Filtering Lag Time (TF): Assumes TF = 0 minutes. After a new range-checked value is determined, it is transferred directly to the PVAUTO value.

2. Overview of the APM/HPM I/O Simulator

2.2. Simulated IOP Functions

Figure 4 Overview of Analog Input Simulator

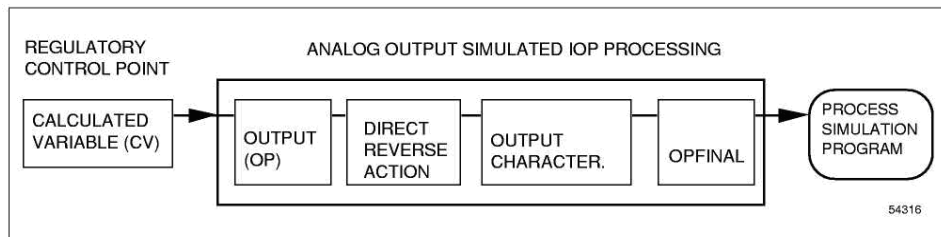


Analog output module

The analog output modules have the following functions simulated:

- Mode, initialization, and full point form functions.
- Output Direct/Reverse action (OPTDIR).
- Output characterization (OPFINAL).

Figure 5 Overview of Analog Output Simulator



Support for time-based functions

In the HPM I/O simulator the following time-based functions may not execute faithfully within their exact real-time specifications:

- DI latched input processing
- DI PV change delay and alarm delay processing
- DO Pulse-Width Modulation (PWM) and one-shot processing.



ATTENTION

The time-based functions are applicable only to the HPM I/O Simulator. The APM I/O Simulator does not support time-based functions.

Digital input module

The digital input modules have the following functions simulated:

- Operator or program is permitted to make stores to the PVRAW value.
- Alarms are detected and processed for all IOP alarms (OFFNORML). This includes alarm priority, alarm enabling/disabling, and contact-cutout functions.
- PV Event Reporting is supported, including Sequence of Events (SOE).
- PVSOURCE selection is supported.
- When a digital input point is configured as an accumulation type, the accumulated value (AV) can be set by the operator or program.
- Input direction (DIRECT/REVERSE) selection is supported.
- For accumulation type digital input points, the point processing related to the AV target value (AVTV) and overflow is executed whenever an operator or program writes to the AV value.



ATTENTION

When a DI status type point whose ALMOPT=OFFNORML or ALMOPT=CHNGOFST and \$DLYOPT=PMOFFDLY is loaded onto the HPM I/O simulator, an error appears for DLYTIME parameter. Therefore, it is recommended to build a point and then use DATACHNG or alter parameters to change the value of DLYTIME to the desired value.

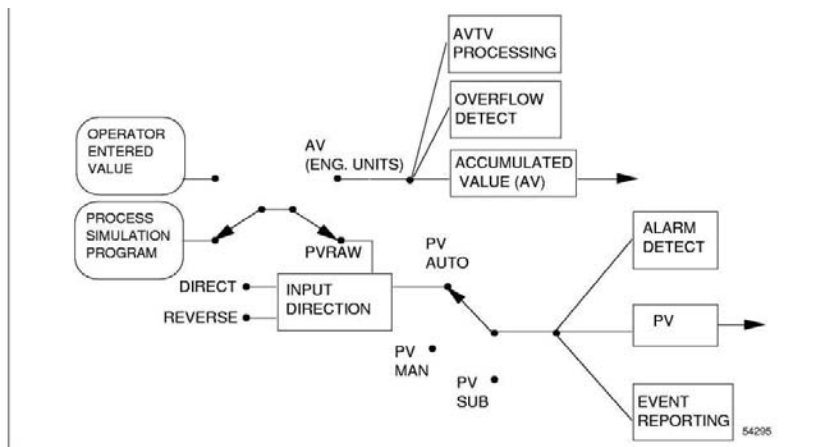
2. Overview of the APM/HPM I/O Simulator

2.2. Simulated IOP Functions

The following functions are not simulated:

- Debounce and the PV change delay filters (DEBOUNCE, PVCHGDLY) can be configured, but both simulators do not support these functions.
- DI latch function can be configured, but both simulators do not support these functions.
- SOE 1ms resolution.

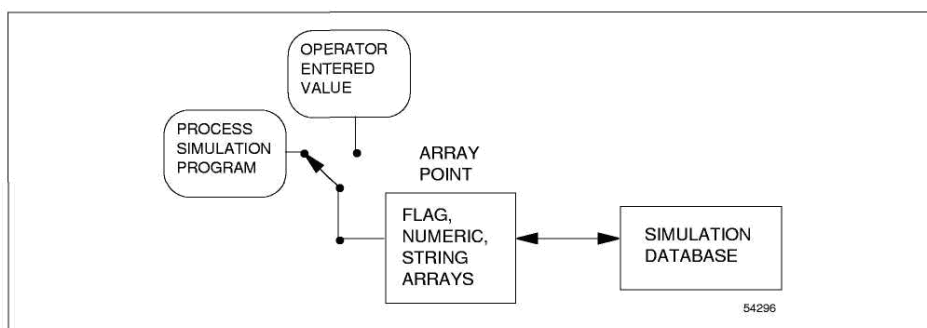
Figure 6 Overview of Digital Input Simulation



Serial interface module

Simulation software permits read and write access to the I/O simulation database for an Array point.

Figure 7 Overview of Serial Input Simulation



Digital output module

Table 1 Simulated/Non Simulated Functions

DO	Simulated Functions	Non Simulated Functions
APM	Digital output status, except OFFPULSE/ONPULSE, can be configured.	- OFFPULSE/ONPULSE outputs - Pulse-Width Modulated (PWM) digital outputs
HPM	Status and Pulse-Width Modulated (PWM) digital output. OPFINAL timing for pulse and PWM outputs.	



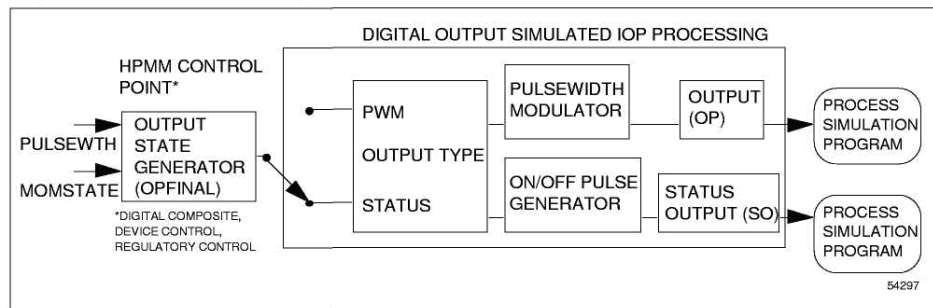
ATTENTION

In an HPM IO simulator, the timing of the DO pulses is not accurate. The actual timing of the pulses can be a second less than the configured value of PULSEWTH.

2. Overview of the APM/HPM I/O Simulator

2.2. Simulated IOP Functions

Figure 8 Overview of Digital Output Simulation



Pulse input module

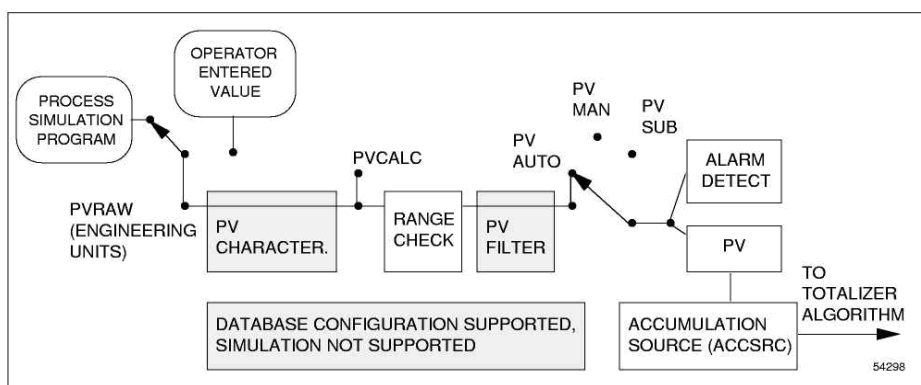
When a pulse input module is in the simulation mode, the accumulation source (ACCSRC) is automatically set to PV. The totalizer algorithm then uses the PV instead of delta AV as the input source.

- Operator or program is permitted to make stores to the PVRAW value. When the APM or HPM is in the simulation mode, PVRAW is in engineering units.
- PVRAW value may be set to NaN (Not a number, —) to simulate a sensor or a transmitter input failure.
- Alarms are detected and processed for the following IOP alarms — PVHI, PVLO, PVHH, PVLL, and BADPV. This includes alarm priority, alarm enabling/disabling, and contact cutout functions. (Although database configuration is supported, PVROCP or PVROCN are not supported.)
- PV range checking and PVSOURCE selection functions are supported.

The following functions are not simulated:

- PV Characterization: As a user-written process simulation program (that is, I/O stimulation software) normally stores to the PVRAW value in engineering units, the PV characterization is not needed. When the PVRAW value is stored, the value is transferred directly to the PVCALC value.
- PV Filtering Lag Time (TF): This assumes TF = 0 minutes. After a new range-checked value is determined, it is transferred directly to the PVAUTO value.

Figure 9 Overview of Pulse Input Simulation



IOP configuration limits for APM I/O simulator

The I/O simulator maintains the IOP database as part of the APMM's communication processor memory. The configuration constraints, summarized in Table 2, are dependent on either of the following:

- The maximum of any one IOP type configuration limit, or
- The IOP configuration mix has resulted in using the amount of available memory. Each IOP has a scaling factor or weight for memory it requires; the total for all IOPs must not exceed 120.

Table 2 APM Configuration Limits

IOP Type	Maximum per IOP Type	Memory Factor for each IOP
DI	30	4
DO	40	2
DO32	40	3
DISOE	30	4
HLAI	24	5
LLAI-8	40	3

2. Overview of the APM/HPM I/O Simulator

2.2. Simulated IOP Functions

IOP Type	Maximum per IOP Type	Memory Factor for each IOP
LLMUX	13	9
SI	13	9
PI	40	3
STI	24	5
AO	40	2
AO16	40	3

Example Calculation

A configuration of 27 AOs and 13 HLAI results in the following:

$$(27 \times 2 \text{ weight of AO}) + (13 \times 5 \text{ weight of HLAI}) = 119.$$

This IOP configuration mix is less than the total memory constraint of 120. Hence, a maximum of 40 primary IOPs can be configured.

A configuration of 8 SIs, 8 DIs, and 8 DOs results in the following:

$$(8 \times 9 \text{ wt of SI}) + (8 \times 4 \text{ wt of DI}) + (8 \times 2 \text{ wt of DO}) = 120.$$

This IOP configuration reaches the total memory constraint of 120, even though the maximum of 40 IOPs is not configured.

IOP configuration limits for HPM

The HPM does not have any IOP configuration limits. The full complement of 40 IOPs can be configured.

Hard failure simulation

A hard failure simulation function is provided through setting the hard failure status of an IOP to any hard failure code. You can access the hard failure codes through a schematic or program, using the following command –

`$NMnnBxx.FAILCODE(i),`

where nn=UCN network number, xx=APM or HPM address number, i = IOP number

After a hard failure is set, a non-redundant IOP acts as if it were a physical non-redundant IOP. Question marks are displayed on operator displays and access to the PV results in an error code. When a hard failure is simulated in the Primary of a redundant IOP pair, both the Primary and the Secondary IOPs go to a “No Response” state. When a hard failure is simulated in the Secondary of a redundant IOP pair, only the Secondary IOP goes to a “No Response” state.

To recover from a simulated IOP hard failure, you can clear the previous hard failure code (FAILCODE) by setting it to UNKNOWN. The IOP returns to the “IDLE” state.



ATTENTION

When there is a hard failure of Primary HPM, the Secondary HPM takes over as Primary, and goes to the Idle state. Since the Secondary HPM does not have a database to run on, it displays the DB Invalid warning when trying to bring up the Secondary HPM to RUN State. Validate the database and then change the state to RUN state.

Example of Hard Failure simulation

The parameter for an IOP's last hard fail status, IOMLHFST, lists the errors for which the FAILCODE can be set. For example, the power down (no response) hard failure status is "POWERDWN." The following is an example for simulation:

```
SET    $NM03B09 . FAILCODE(1) = POWERDWN
```

RESULT: The status of IOP module #1 goes to the No Response (NR) state.

To remove the error, the FAILCODE is set to UNKNOWN:

```
SET    $NM03B09 . FAILCODE(1) = UNKNOWN
```

RESULT: The status of IOP module #1 goes to the IDLE state.

2.3 Physical IOP Functions

Overview

Because IOP functions are simulated by the APM or HPM communication processor in a simulation mode, all the IOP data and status information that is accessed, represents the simulated data and the status.

Physical IOPs are inaccessible

If physical IOPs are installed, the database and status of the physical IOPs are not accessible in the simulation mode. Because the physical I/O is not accessible, field wiring checkout in parallel with control strategy checkout is not supported in the same APM or HPM. Additionally, any physical IOP failure is not detected or reported when either the APM or the HPM is in the simulation mode; therefore, the IOPs are left in the “FAILURE” state.

Accessing the IOPs after simulation

To access physical IOPs after a simulation is complete, the APM or HPM must be reloaded with the on-process personality. The APM or HPM must go to the “ALIVE” state to load the on-process personality after a simulation.

Physical output action during simulation

For all the physical output modules (AO and DO) present in either the APM or the HPM I/O simulator, the time-out gate function on the IOPs expire. When the time-out gate expires, the physical output type processors assume they have lost communication with the APMM or HPMM. As a result, the physical outputs are set to the user-configured Fail State—which can be “either hold the last output value” or “go to an unpowered state.”

SI IOP simulation in HPM

The SI IOPs are not simulated if they are physically present and configured when the HPM is in the simulation mode. The IOL communication occurs for these IOPs. The simulation actions are applicable only to those configured SI slots that do not have physical SI IOPs present.

The support mode of the IOP, real or simulation must be done during configuration. Insertion or removal of physical IOP during normal simulation operation does not change the support mode of SI IOP.

2. Overview of the APM/HPM I/O Simulator

2.3. Physical IOP Functions

No change occurs if the SI IOP card is installed when an HPM is in a simulation mode. If the SI IOP card is removed when it is operating in the real mode, the Detail Display shows “NR” (no response). If the IOP is re-inserted, it continues to function in the real mode.

The SI IOPs function in real mode when the IOPs are physically present in the HPM. Under this mode the SI data is accessed directly from the SI card. SI IOP without hardware is simulated as usual.

2.4 Redundancy Function

Redundancy Configuration is supported

The I/O Simulator supports configuration of the APMM or the HPMM and I/O redundancy. Checkpoints of databases with redundant configurations can be transported to and from the target APM or HPM. The target APM or the HPM can be an APM or an HPM designated for on-process control.

APMM or HPMM redundancy behavior

The APMM or the HPMM redundancy behavior is summarized as follows:

- The Primary and the Secondary APMM or HPMM can be loaded with the simulation personalities. When this occurs, the simulation APMM or HPMMs appear as physical APMM or HPMMs in the status displays. Operational commands such as IDLE, STARTUP, and SHUTDOWN are permitted.
- In a redundant APMM or HPMM with I/O simulation personality loaded, the secondary is in the backup state.
- After a failover or swap, all IOPs go to the “Idle” state, with DB_INVALID displayed. This means that the IOP database has been defaulted (cleared). In addition, the APMM or the HPMM goes to the “Idle” state and the control database is defaulted (cleared) to no points.
- Redundant APMM or HPMM cannot be loaded with both the on-process and the I/O simulator personalities at the same time. That is, the on-process and the I/O simulator personalities cannot coexist in a redundant APMM or HPMM. In such a case, the secondary goes to the fail state.



ATTENTION

When redundant APMMs or redundant HPMMs are swapped (that is, the Secondary becomes the Primary), all IOP databases are cleared. To avoid unintended loss of the IOP databases, leave the Secondary in the “ALIVE” state when an APM or an HPM is loaded with the simulation personality.

IOP redundancy behavior

The IOP redundancy behavior is summarized as follows:

- IOPs configured as redundant have a detailed display status of OK/BKP.
- Redundant IOPs ignore swap commands from an operator and do not failover if a simulated hard failure is requested.

The effect of transition into a simulation mode for redundant configuration is described in the Simulation indication (see tabs Install and Operate).



ATTENTION

In a redundant configuration, the Simulator does not support the following IOL cable operations:

- Selecting cables – A and B
 - Swapping cables – A and B
-

Key differences between On-process APM/HPM and APM/HPM simulator

The following are the key differences between the on-process APM/HPM and the APM/HPM I/O simulator:

On-process APM/HPM functions	Simulator functions	Applicable to...
If OP is set when an AO point is a full point, the OP does not change when the AO point is made a component point.	If OP is set when an AO point is a full point, the OP is reset to zero when the AO point is made a component point.	APM/HPM I/O simulator
Node specific configuration is allowed only when a box is in IDLE state.	Node specific configuration is allowed even when a box is in OK state.	APM/HPM I/O simulator

2. Overview of the APM/HPM I/O Simulator

2.4. Redundancy Function

On-process APM/HPM functions	Simulator functions	Applicable to...
For a DI point, the following parameters can be modified only when the PTEXECST is INACTIVE: • ALMOPT • INPTDIR	For a DI point, the following points can be modified even when PTEXECST is ACTIVE: • ALMOPT • INPTDIR	HPM I/O simulator
For an AO point, OPTDIR can be modified only when the PTEXECST is INACTIVE.	For an AO point, OPTDIR can be modified even when the PTEXECST is ACTIVE.	HPM I/O simulator
For an AI point, the following parameters can be modified only when the PTEXECST is INACTIVE: • Range limit • INPTDIR • SLWSRCID	For an AI point, the following parameters can be modified even when PTEXECST is ACTIVE: • Range limit • INPTDIR • SLWSRCID	HPM I/O simulator

3. Installation and Operation

3.1 Before you begin

Introduction

Review this section to ensure that you have made backup copies of the system checkpoint and are aware of any additional considerations.

File Naming

The APM and HPM simulation personalities are contained on a cartridge/CD. Although these personalities can be copied to and loaded from your HM, load the simulation personality from removable media only. The reason for loading from removable media is to keep the simulation personalities off the network, where an operator could inadvertently load them into an operating system controlling the process. The filename for the simulation personalities are as follows:

Table 3 Simulation Personalities

	DIR	Filename	Extension
APM	&UCN	APMCOMS1	.PI
	&UCN	APMCOMS2	.PI
	&UCN	APMCTLS1	.PI
	&UCN	APMCTLS2	.PI
HPM	&UCN	HPCMHS1	.PI
	&UCN	HPCMHS2	.PI
	&UCN	HPCTHS1	.PI
	&UCN	HPCTHS2	.PI

The letter "S" in the file name indicates that a simulation personality is loaded in the HPM or the APM.

Example of backing up media

Prepare a formatted cartridge/CD for backing up the simulation programs. Copy the entire contents of the distribution media to the blank cartridge/CD.

3. Installation and Operation

3.1. Before you begin

CP \$F1>&UCN>*. * \$F2>&UCN>= -D

\$F1 is the drive you are copying from and \$F2 is the drive you are copying to.

Operate with the backup media

Store the distribution media in a safe place. Use the backup copy for installing and operating the APM or the HPM simulator.

Backup your system data

Backup network resident checkpoints on another cartridge/CD, in case the data is unintentionally overwritten. Place this backup in a safe place and mark it appropriately. List the files on the cartridge/CD to ensure proper backup.

3.2 Installing the I/O Simulator

Introduction

Load the simulation personality to the APM or the HPM as you normally would, using the same UCN Status Displays for the on control personality. In R500 and later, the UCN Status Displays provide a SIMULATOR LOAD target for loading the I/O simulator. The simulator personalities can be copied to and loaded from the History Module. For the safety of your process, it is suggested that you load the personalities from the R500 and later APM/HPM I/O Simulator media.

Before you begin

CAUTION	The following usage guidelines are applicable to the APM/HPM I/O simulation personalities in systems connected to the process or to a simulation.
----------------	---

- Do not operate the APM or the HPM I/O simulator in an on-process system. Do not load the simulator personality into the APM or HPM when operating on-process system. When the simulation personality is loaded into the APM/HPM, the physical IOPs become inaccessible.
- Automatic checkpointing is disabled in the simulation mode. Automatic checkpointing is disabled because an automatic checkpoint is made to the History Module overwrites the previous checkpoint (which may have been a checkpoint intended for control, training, or strategy checkout).
- Honeywell recommends that the simulation mode checkpoints be made to removable media. Checkpoints to removable media let you set up a variety of simulation states, which are useful for operator training and system checkout.

Systems not connected to the process

For systems typically connected to a process (and not used for operator training and control strategy checkout systems), the I/O simulator personality can be copied into the appropriate HM system volume. However, it is easier to manage and maintain control of your simulation and operating personalities when you load them from removable media.

3. Installation and Operation

3.2. Installing the I/O Simulator

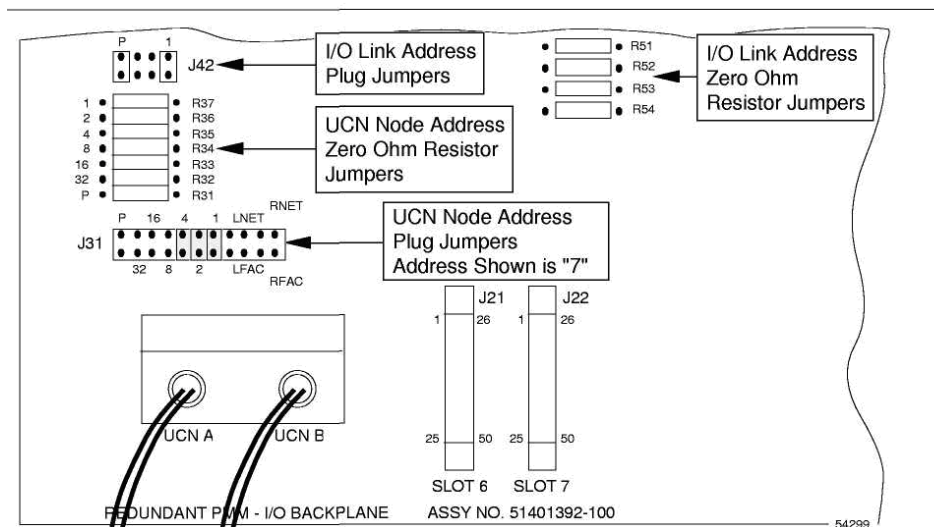
Changing the APM or the HPM address

If the target APM or the target HPM (that is, an APM or an HPM intended for process control, which can be on a separate network) has a different node address than the simulation APM or the simulation HPM, the hardware address of the simulation APM or the simulation HPM can be reset to match the target APM or the HPM.

Example APM address

The location of the APM address settings is given in Figure 10, while a full description is available in Honeywell technical publication PM13-620, *Process Manager/Advanced Process Manager Service manual*.

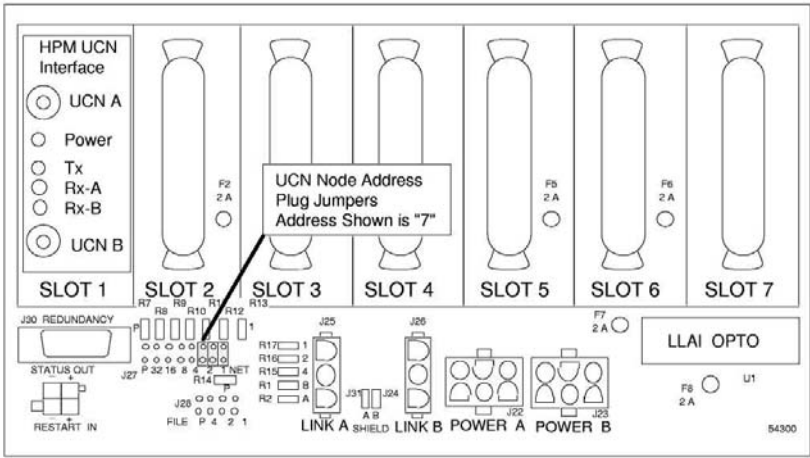
Figure 10 Address Jumper Location



Example HPM address

The location of the HPM address settings is given in Figure 11, while a full description is available in Honeywell technical publication HP13-610a, *High-Performance Process Manager Service manual*.

Figure 11 HPM Address Jumper Location



Primary and secondary addresses

Recall that the Primary and the Secondary APM/HPM addresses are pinned to the same number when the APMM or the HPMM redundancy is file-to-file. For example, if the primary APM or the primary HPM address is set to "7," the secondary APM or the secondary HPM is also set to "7."

NIM Address and UCN recommendation

The NIM address settings and the UCN Network numbers in the off-process simulation system and target system must be set to the same address. This enables checkpoint transportability between systems.

Loading the program

Prior knowledge of loading APM or HPM personality software is necessary for loading the simulator personality. For further explanation and procedures, see *Process Operations Manual*, SW11-615.



ATTENTION

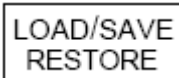
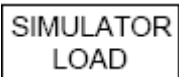
If the target APM or the HPM for the simulation personality is connected to process control hardware and you are not sure of FAILOPT action (unpowered or hold), reconfirm the configured FAILOPT for output type IOPs prior to loading the simulation personality.

3. Installation and Operation

3.2. Installing the I/O Simulator

The steps in Table 4 show you one of the ways to load the APM or the HPM's simulator program. The display results are shown in Figure 12.

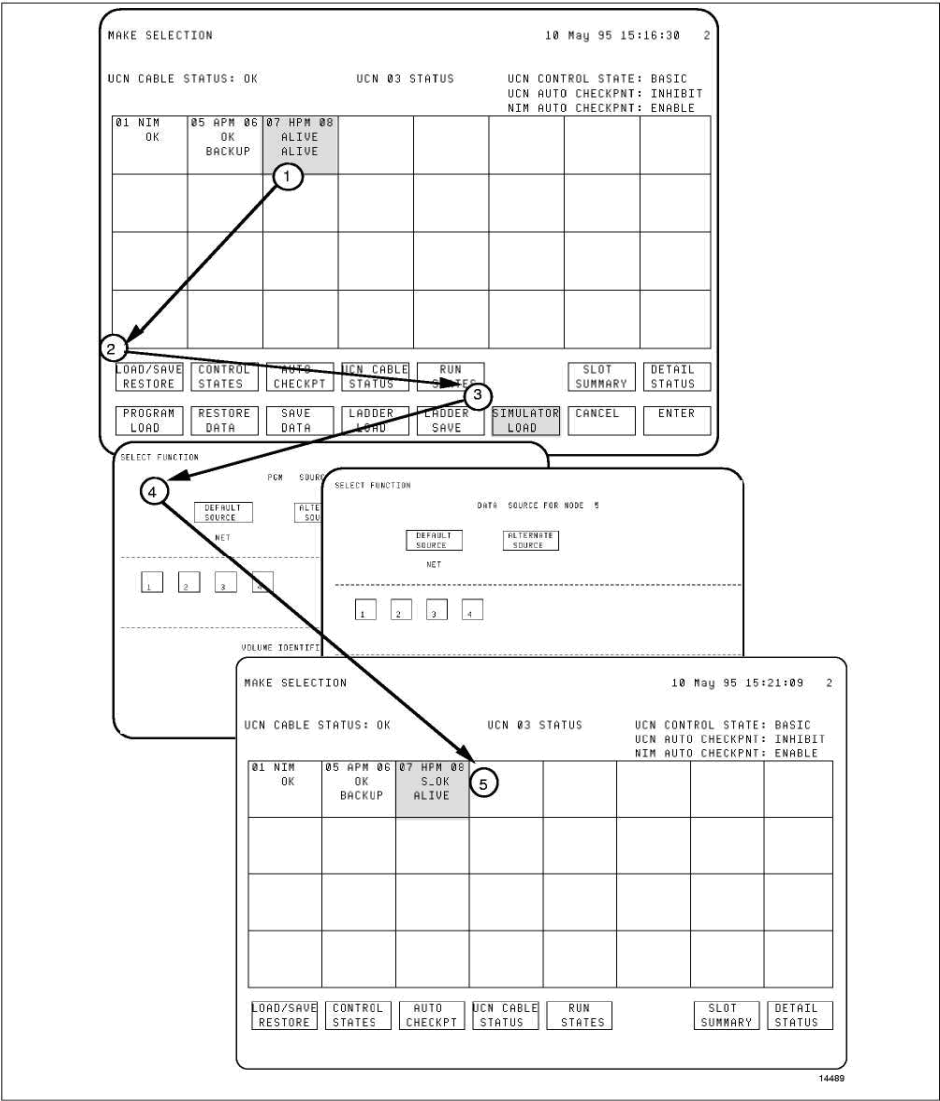
Table 4 Loading APM/HPM Simulator

Step	Action
1	From the UCN Status Display, select the target representing the primary APM or HPM you wish to load.
	 The APM or HPM and its redundant partner must be shutdown and in the ALIVE state to load the simulation personality.
2	Select the  target.
3	Select the  target, press ENTER.
4	Select the  targets if your APM or HPM simulator personalities and checkpoints are on a removable media.
5	Select the  target. Repeat steps 1 through 4 for a redundant APM or HPM. Response: After the loading is completed, the primary APM or the primary HPM appears in the simulation Idle (S_IDLE) state.
	 If the APM or HPM is loaded with just the simulation personality and no checkpoint, the APM or HPM has no control or I/O database. You can load an APM or HPM checkpoint using RESTORE DATA or load new points from the Data Entity Builder while the APM or HPM is in S_IDLE. After issuing startup commands to the APM or the HPM, the status is displayed as S_OK/ALIVE.

Loading the program

Figure 12 is an example of the expected display results.

Figure 12 Program Load Example



3.3 Simulation Group and Detail Indications

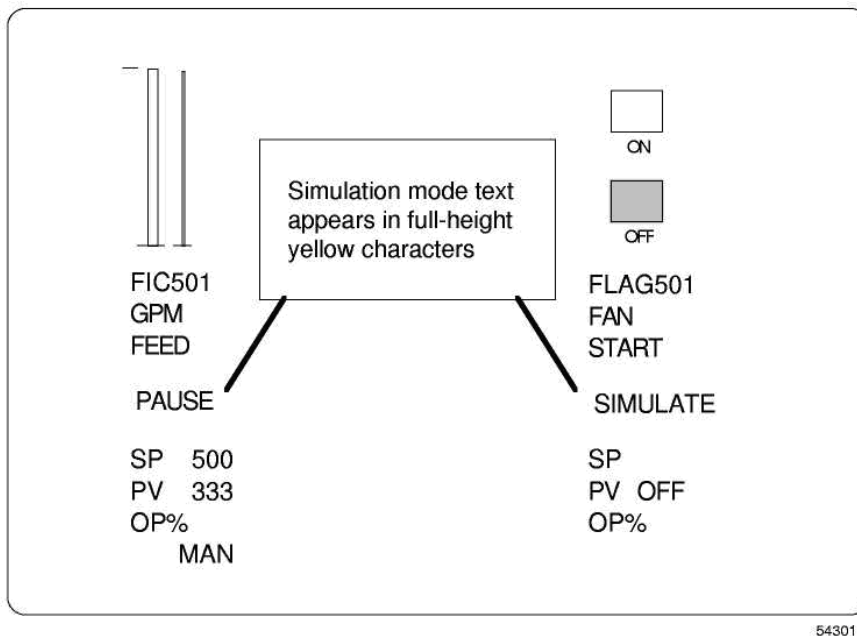
Simulation indication

A standard indication of the simulation mode is provided on the node status displays, while per-point indication of the simulation mode is based on your configuration of the simulation display indicator parameter - DISP_SIM, specified during node specific configuration (see Figure 24).

Display Text Examples

After the I/O simulation personality is loaded and the points are operational, the optional text appears for simulation points in group and detail displays as shown in Figure 13.

Figure 13 Simulation Text



Display text action

The display text action described in Table 5 is based on the parameter SIM_TXT, which resides in the following R500 and later versions of the APM or the HPM simulation mode points—AI, AO, DI, DO, RegPV, RegCtl, DevCtl, Logic, Array, Flag, Numeric, DigComp, and Timer.

Table 5 Simulation Text Indicator

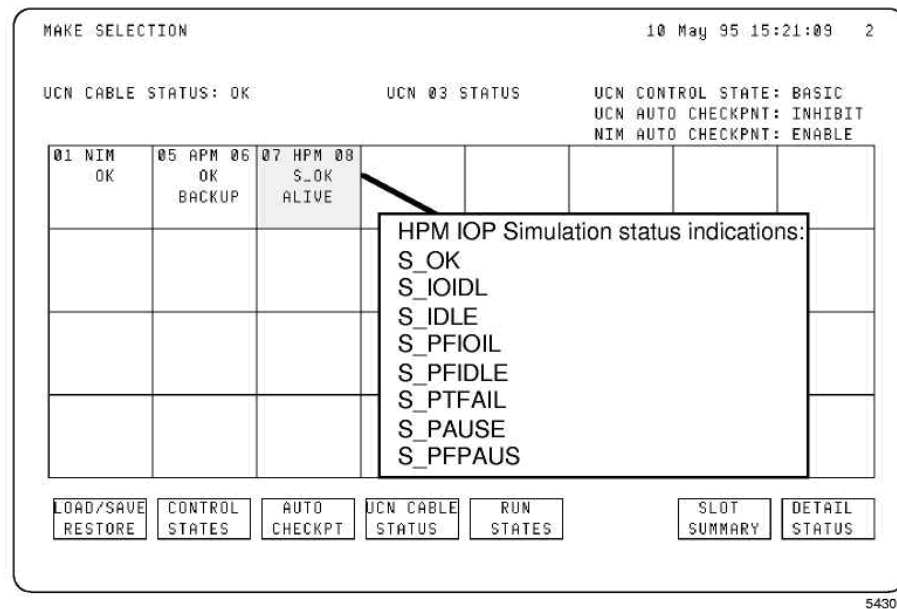
Text	Description
SIMULATE	The point is in a simulation state, the APM or HPM is in a simulation status other than S_PAUSE or S_PFPAUS, and the simulation display indicator parameter DISP_SIM, is configured as "on."
PAUSE	The APM or HPM is in a simulation status S_PAUSE or S_PFPAUS. An APM or an HPM in pause (S_PAUSE or S PFPAUS) always shows PAUSE indication, regardless of DISP_SIM selection.

3.4 Simulation Status Indications

UCN stimulation status indications

An example display with an HPM in the simulation mode appears in Figure 14.

Figure 14 Overall UCN Simulation Status



Description of the simulation states

The simulation states are similar to the on-process states, except that the simulation states have an "S_" prefixed to the state.

Table 6 APM or HPM Simulation State Indicators, UCN Status Display

State	Description	Possible transition states
S_OK	The simulation is OK. The IOPs and the APMM or the HPMM are processing points and functioning normally.	S_OK to S_PAUSE S_OK to S_IDLE
S_IIDL	An IOP is in the IDLE state. The APM or the HPM is in the simulation mode.	
S_IDLE	The APM or the HPM is in the IDLE simulation state and there is no point processing .	S_IDLE to S_OK (cold or warm restart) S_IDLE to ALIVE (database cleared)
S_PFIOL	A partial failure has occurred in one or more IOPs, which are in the IDLE state.	
S_PFIDLE	A partial failure has occurred in the APMM or the HPMM that is in the IDLE state.	
S_PTFAIL	A partial failure has occurred in the simulation APMM or HPMM.	
S_PAUSE	The APM or the HPM is in the PAUSE simulation mode and no point processing occurs in this state.	S_PAUSE to S_OK (hot restart) S_PAUSE to S_IDLE S_PAUSE to ALIVE (database cleared)
S_PFPAUS	A partial failure has occurred when the simulation is in the pause state.	

HPM simulation status indication

An example display with an HPM in the simulation mode is shown in Figure 15.

3. Installation and Operation

3.4. Simulation Status Indications

Figure 15 Overall HPM Simulation Status

MAKE SELECTION 10 May 95 15:23:50 2

HPM AUTO CHECKPNT: INHIBIT
IOL PERIODIC SWAP: ENABLE

HPM 07 STATUS/UCN 03 HPM CONTROL STATE :BASIC
HPMM 07 P HPMM 08 S UCN CBL STS HPMM 07: B/A
S_OK ALIVE HPMM 08: B/A

WRITE LOCKOUT : OFF

01 HLAI OK /BKP	02 AO OK	03 AO OK	04 DO OK	05 DI OK	06 LLAI OK	07 SI OK	08
09	10	11	12	13	14	15	16
17	18	19	20				
25	26	27	28				
33	34	35	36				

HPM IOP Simulation status indications:
S_IDLE
S_IDLESF
S_OK
S_SFFAIL
S_PAUSE
S_PAUSSF

LOAD/SAVE RESTORE CONTROL STATES AUTO CHECKPT IOL CABLE COMMANDS RUN STATES SLOT SUMMARY DETAIL STATUS

54303

UCN status simulation indication

The UCN simulation states are similar to the on-process states, except that the simulation states have an "S_" prefixed to the state.

Table 7 Simulation State Indicators (APM or HPM Status Display)

State	Description
S_OK	Simulator personality is running and point processing is occurring.
S_IDLE	Simulator is in IDLE state.
S_IDLESF	Simulator is in IDLE and a soft failure was detected.

State	Description
S_SFFAIL	Simulator is running, however a soft failure was detected.
S_PAUSE	Simulator is in a PAUSE state.
S_PAUSSF	Simulator is in a PAUSE state and a soft failure was detected.

Mismatched personalities

CAUTION

If a redundant APMM or HPMM partner is loaded with the simulation personality and the other partner contains the on-process personality (or conversely), the second node loaded will fail. A load of the secondary of an on-process node with the simulation personality is intentionally prevented for security reasons.



ATTENTION

When the APMMs or the HPMMs are swapped, all IO databases are cleared.

Simulated redundant IOP status

The I/O simulator supports I/O redundancy configuration, but not IOP redundancy operation. For any configured redundant IOP pairs, the "B" appears as a backup (BKP), as shown in Figure 16.

3. Installation and Operation

3.4. Simulation Status Indications

Figure 16 Simulated redundant IOP status – Example

MAKE SELECTION
10 May 95 15:23:50 2

HPM AUTO CHECKPNT: INHIBIT
IOL PERIODIC SWAP: ENABLE

WRITE LOCKOUT : OFF

HPM 07 STATUS/UCN 03

HPMM 07 P	HPMM 08 S
S_OK	ALIVE

HPM CONTROL STATE :BASIC

UCN CBL STS HPMM 07: B/A
HPMM 08: B/A

01 HLAI OK /BKP	02 AO OK	03 AO OK	04 DO OK	05 DI OK	06 LLAI OK	07 SI OK	08
09	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40

LOAD/SAVE
RESTORE

CONTROL
STATES

AUTO
CHECKPT

IOL CABLE
COMMANDS

STATES

SUMMARY

STATUS

HPM IOP Simulation status
on redundancy IOP pairs

IOP diagnostic simulator message

The simulator status is also shown on the associated UCN status and the APM or the HPM node status displays as shown in Figure 17. The following status information is available on their diagnostic displays:

- The message IOP in simulation is shown in red.
- The firmware revision field indicates the simulator software version.

Figure 17 IOP diagnostic display – Example

MAKE SELECTION 10 May 95 15:25:33 2

HPM IOP Simulation indication

HPM Simulator Software version

UCN 3 HPM 7 MODULE 1 PHYS ADDR 160 FILE 3 CARD 1
HLAI PRIMARY IOP_A
STS OK ACTUAL TYPE HLA1 FTA Connect CONN_A With Bias Yes
Status: ** IOP in Simulation **
HARDWARE REV 00 FIRMWARE REV 5.0 LAST HARD FAILURE POWERDOWN
DESCRIPTION: Iosimul
ACTIVE CABLE CHANNELA CHN A ERR : 0 CHN A SIL : 0
LAST COMM ERR NONE CHN B ERR : 0 CHN B SIL : 0
RESET MODULE
COMM ERRORS

FTA PRESENT NO STANDBY MANUAL
COLD POWERUP LAST EXECUTED NO SWITCHED ACTIVE
Secondary IOP Status

Box Soft Failures Slot Soft Failures Slots with Soft Failures

CALIBRATION NOT PERMITTED, MODULE NOT IN IDLE STATE

54305

The APM display is similar to the HPM display.

APM node diagnostic indication

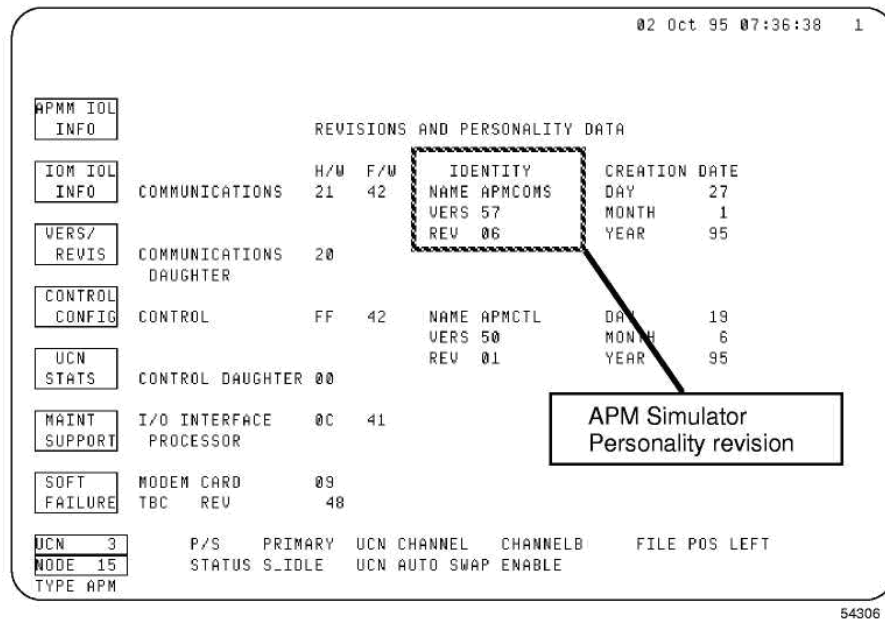
In addition to the simulation status indicator on the IOP diagnostic display, the APM node diagnostic display shown in Figure 18 can be checked to verify that the I/O simulator personality is loaded.

In the simulation mode, the communication personality version is "5x" or later (where "xx" indicates a revision series).

3. Installation and Operation

3.4. Simulation Status Indications

Figure 18 Revision status – Example



HPM node diagnostic indication

In the simulation mode, the communication personality version is "5x" or later (where "x" indicates a revision series) as shown in Figure 19.

Figure 19 Revision status – Example

10 May 95 15:26:45 2

HPMM IOL INFO	REVISIONS AND PERSONALITY DATA					
	OVERALL SW VERS 58 REV 17					
IOM IOL INFO	H/W F/W		IDENTITY		CREATION DATE	
VERS/ REVIS	COMMUNICATIONS	A DF	NAME HPCMHS	DAY	27	
			VERS 57	MONTH	2	
			REV 02	YEAR	95	
			LOADED PERSONALITY: I/O SIMULATION			
CONTROL CONFIG	CONTROL		NAME HPCTHP	DAY	24	
			VERS 58	MONTH	3	
			REV 12	YEAR	95	
			LOADED PERSONALITY: ON-PROCESS			
UCN STATS						
MAINT SUPPORT	I/O INTERFACE	A	VERS 58	DAY	18	
			REV 00	MONTH	8	
				YEAR	94	
SOFT FAILURE	UCN INTERFACE	?				
	TBC REV	I				
UCN 3	P/S	PRIMARY	UCN CHANNE	FILE_1		
MODE 7	STATUS S_OK		UCN AUTO S			
TYPE HPM						

HPM Simulator
Personality revision

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3. Installation and Operation

3.5. Pausing and Resuming the Simulation

3.5 Pausing and Resuming the Simulation

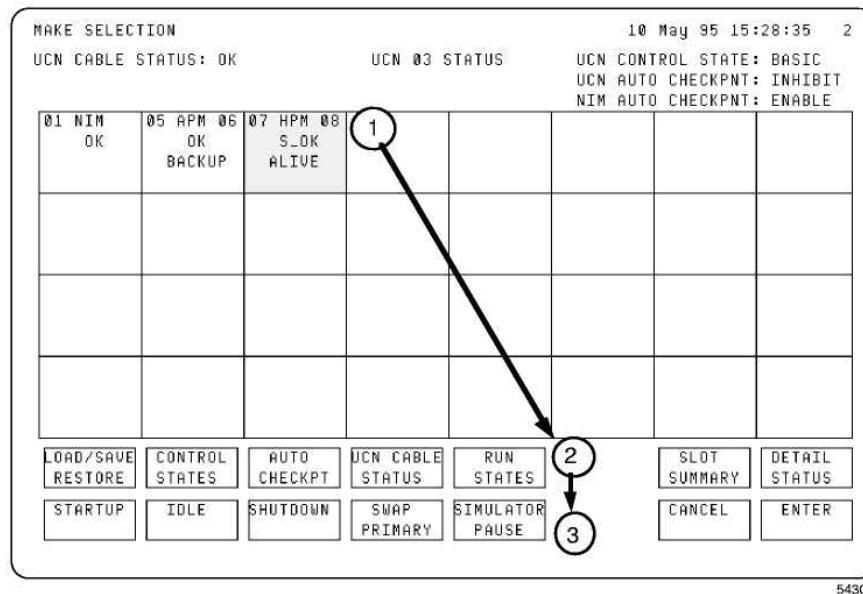
Introduction

You can pause a simulation to check conditions, checkpoint database, and then resume the simulation.

Pause

You can pause a simulation in progress when the APM or the HPM is in S_OK or S_PTFAIL state. In the Pause state, the simulation PVRAW and the OP cannot be changed except through a checkpoint restore. Additionally, the control points retain their values. Because the LCN system time clock cannot be frozen, any configured history collection continues in the Pause state.

Figure 20 Pausing a Simulation Mode HPM



Pause procedure

Table 8 Pause APM or HPM Simulation

Step	Action
1	After setting the access key to Engineering, select the primary APM or the primary HPM that is in the S_OK or S_PTFAIL state. See Figure 20.
2	Select the RUN STATES target.
3	Select the SIMULATOR PAUSE target and press [ENTER].



ATTENTION

On the detail and group displays of the APM I/O simulator, the SIMULATOR/PAUSE indicator is updated only when the display is invoked again. For example, when an APM is in the simulation mode, the SIMULATOR/PAUSE indicator shows SIMULATE. If the simulation is paused, the SIMULATOR/PAUSE indicator continues to show SIMULATE. The indicator is updated only when the display is invoked again.

Idle versus Pause

While an IDLE command can be used for interrupting a simulation, the effect of issuing an IDLE command to the simulation mode APM or HPM is to permit some initialization of dynamic values. For example, when issuing a startup command to an APM or HPM in S_IDLE, you need to select whether you want a cold startup or a warm startup.

While a PAUSE command can be used for interrupting a simulation, the effect of issuing a PAUSE command to a simulation mode APM or HPM is to permit a warm restart. For example, on issuing a startup command to an APM or HPM in S_PAUSE, you do not need to select whether you want a cold startup or a warm startup. A warm start occurs after selecting the Resume target.

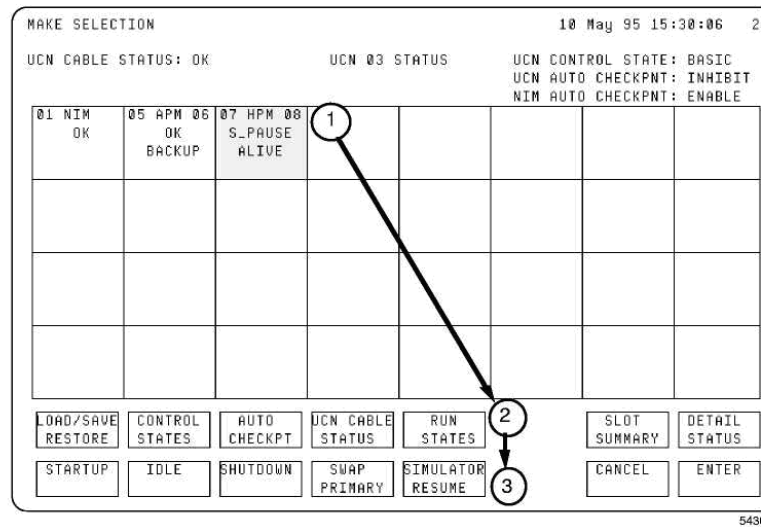
Resume

To resume the simulation, the paused APM or the HPM must be issued a resume command. Internal control dynamics are not lost during the start initialization pass. However, if the APM or HPM is restored with a checkpoint from actual IOPs, the dynamic values (PVRAW, AV, OP) are set to the necessary values. In addition, the CL/APM or the HPM programs are restarted from the CL/APM or the HPM statement in the first sequence Phase and Step. However, the CL/APM or the HPM programs are lost and "NL" is displayed in the Process Module Detail Display if checkpoints are restored during S_PAUSE.

3. Installation and Operation

3.5. Pausing and Resuming the Simulation

Figure 21 Resuming a simulation



Resume procedure

Table 9 Resume APM or HPM Simulation

Step	Action
1	After checking that the access key is set to Engineering, select the target representing the primary APM or the primary HPM in S_PAUSE or S_PPFAUS. See Figure 21.
2	Select the RUN STATES target.
3	Select the SIMULATOR RESUME target and press [ENTER].

Previous Checkpoints

To resume a previously checkpointed simulation, perform the following:

- Pause the simulation APM or HPM
- Restore the checkpoint to be used as starting point for replay
- Start the APM or the HPM with a Resume command.

The simulation resumes using checkpointed PVRAW data with limitations as described previously in the "pause" and "resume" descriptions.

APM or HPM pause behavior

When the APM or HPM goes to the Pause state, the following applies:

- Control execution is suspended.
- No initialization to bad values occurs.
- Parameter access is permitted. However, an exception to this is that the PVRAW and the OP cannot be changed during the Pause state.
- Alarm detection and reporting is suspended. In addition, the Control Processor alarms are not recovered or reissued when the HPM simulator returns to the RUN state.
- Checkpoint loading is permitted.
- Warm restart (that is, use current dynamic values for PVRAW, AV) occurs on resume command invocation.
- When the Simulator is in a Pause state, no change occurs if an IDLE command is attempted.
- When the HPM I/O simulator is in a Pause state, confirmation messages are lost. The confirmation is not displayed when the Simulator resumes its state.

3.6 Building a New Database

Background

When an I/O simulation personality is loaded without a checkpoint, the simulation IOP database is marked "Invalid" for any newly configured IOP. To create a database, the standard procedures for IOP point building can be followed. If an existing database is available, from an on-line system or from a previously saved simulation database, this database can be loaded by placing the APM or HPM into S_IDLE or S_PAUSE and following APM or HPM checkpoint restore procedures (an example appears in Table 10).

Configuring IOPs

The simulated IOPs are configured as part of the APM or the HPM box configuration using the Parameter Entry Display, just as in configuring an actual physical APM or HPM with IOPs. The simulated IOPs may be deleted or changed on-line if the IOP is in an S_IDLE or S_PAUSE state.

Quick Reference List

An overview of the APM and the HPM node configuration procedure is described in this section. Additional details are available in Section 3, Point Building in either the *APM Implementation Guidelines*, AP12-60 or the *HPM Implementation Guidelines*, HP 12-500.

Configuring APM HPM points

The simulated I/O data points are configured using the Parameter Entry Displays, just as in configuring the actual physical APM or HPM with IOPs. Both APM and HPM Point Configuration procedures are described in the *Data Entity Builder* (SW11-611) document.

Simulation Parameters

The following parameters have been added to the APM and HPM in support of the R500 I/O Simulation Personalities. You may find these parameters useful in monitoring the I/O simulation personality. Additional details about these parameters can be found in the *XPM09-660 PM Family Parameter Reference Dictionary*.

Table 10 Simulation Parameter Support

Parameter	Description	Type of support
NODECMD	Node command	Enumerations include pause and resume.
NODESTS	Node status	Enumerations for simulation states that appear in the UCN Status display.
PMMSTS	Primary APM or primary HPM status	Enumerations for simulation states that appear in APM or HPM Status display.
SIM_TXT	Simulation indicator	Eight character string (simulate or pause), view only.
DISP_SIM	Simulation indicator option	Option is on or off to display "simulate" text in group and detail displays.

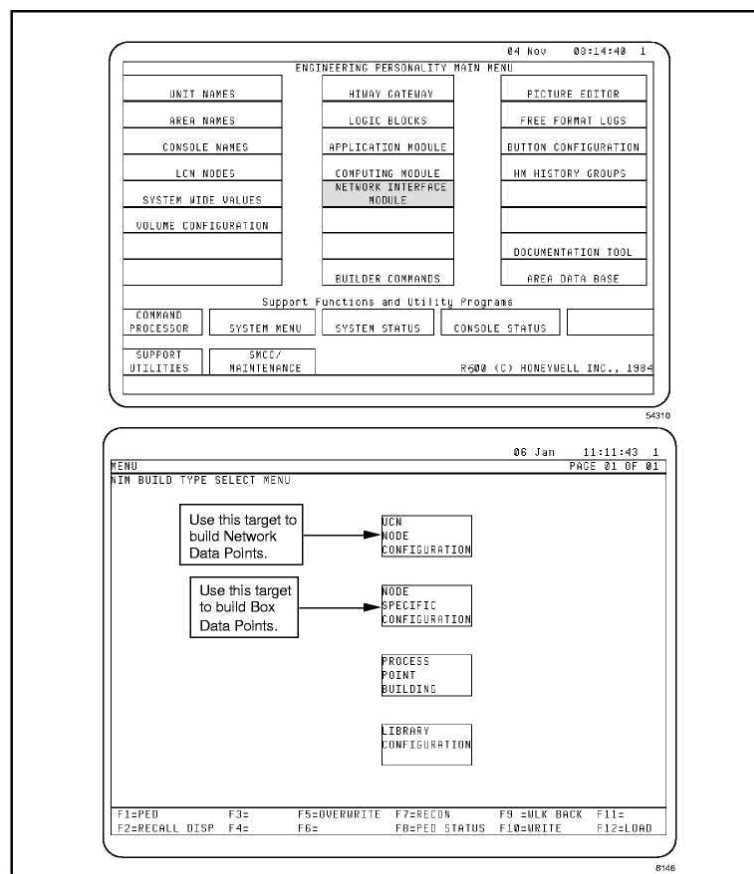
3.7 Configuration Overview

Introduction

Recall that the APM and the HPM node specific configurations are performed by

selecting the **NETWORK INTERFACE MODULE** and **NODE SPECIFIC CONFIGURATION** targets from the menus seen in Figure 22.

Figure 22 Configuration Menus



Example Page 1

Node specific configuration is required for those IOPs you want to simulate.

Figure 23 Example Page 1

The screenshot shows a terminal window titled "PED >>>>> POINT:\$NM UNIT: UNENTERED PAGE 01 OF 01". The date and time "10 May 95 15:33:19" are in the top right corner. The main title is "NODE-SPECIFIC CONFIGURATION". Below it, there are three rows of configuration options: "NETWORK NUMBER (NTWKNUM)" with a checkbox, "NODE NUMBER (NODENUM)" with a checkbox, and "NODE TYPE (NODETYP)" with four radio button options: "PM", "APM", "HPM", "LM", and "SM". The "PM" option is currently selected. At the bottom of the screen, there is a row of function key definitions: F1=PED, F2=RECALL DISP, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WLK BACK, F10=WRITE, F11=, and F12=LOAD.

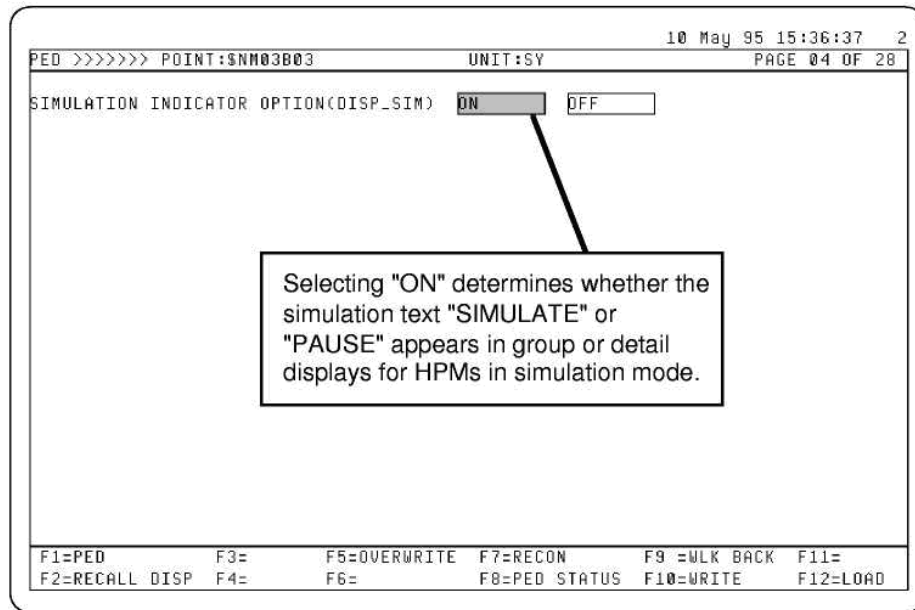
```
PED >>>>> POINT:$NM UNIT: UNENTERED PAGE 01 OF 01
10 May 95 15:33:19 2
NODE-SPECIFIC CONFIGURATION
NETWORK NUMBER (NTWKNUM) ☐
NODE NUMBER (NODENUM) ☐
NODE TYPE (NODETYP) PM APM HPM
LM SM
F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD
```

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Example Page 3

An example page 3 is shown in Figure 24. When the simulation indicator option is ON, the text SIMULATE or PAUSE appears in group and detail displays for all points assigned to that HPM (see Figure 13). When the indicator option is OFF, PAUSE appears only when the APM or the HPM is in a simulation mode.

Figure 24 Example Page 3



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Example IOP Page

Note the IOP simulation selections for the following:

- High Level Analog Input (HLAI)
- Smart Transmitter Interface (STIM)
- Pulse Input (PI)
- Low Level Analog Input Multiplexer (LLMUX)
- Low Level Analog Input (LLAI)
- Analog Output (AO and AO_16)
- Digital Input (DI)
- Digital Input Sequence of Events (DISOE)
- Digital Output (DO and DO_32)
- Serial Interface (SI)

Figure 25 Example IOP Page

PED >>>>> POINT:SNM03B03		UNIT:SV		11 Oct 05 11:46:16 8	
IO MODULE CONFIGURATION					
MODULE NUMBER		(MODNUM) 1			
IOM-A FILE NUMBER		(IOMFILEA(1)) 1			
IOM-A CARD NUMBER		(IOMCARD(1)) 03			
MODULE TYPE		(IOMTYPE(1))			
		NONE ☐ HLAI ☐ STIM ☐ PI ☐			
		LLMUX ☐ LLAI ☐ AO ☐			
		DI ☐ DISOE ☐ DO ☐ ☐			
		AO_16 ☐ DO_32 ☐			
		FBUS ☐			
REDUNDANCY OPTION		(IOREDOPT(1))			
		☐ REDUN ☐			
MODULE NUMBER		(MODNUM) 2			
F1=PED		F3=		F5=OVERWRITE	
F2=RECALL DISP		F4=		F6=	
				F7=RECON	
				F8=PED STATUS	
				F9 =VLK BACK	
				F10=WRITE	
				F11=TAB	
				F12=LOAD	



REFERENCE - INTERNAL

Additional detail about any of these configuration choices can be found in the reference documentation listed at the beginning of this publication.



ATTENTION

The IOL cable selection and cable swapping operations are not supported in the APM/HPM I/O simulator.

3.8 Checkpointing Simulation Database Guidelines

Background

Checkpointing operations access either the physical I/O database or the simulation database. With the standard on-process personality loaded, the physical IOP database can be saved or restored.

Checkpointing simulation database

With the I/O simulator personality loaded, the simulated IOP database can be saved or restored. When switching between on-control and simulation, the I/O database must be saved before shutting down the APM or HPM. This is because the Shutdown command causes the database to be cleared (initialized) in an APM or HPM.

Manual /Auto check points

The I/O simulation database can be manually or automatically saved in the simulation mode. Manual save is provided as an operator command. Auto checkpointing, if required, must be re-enabled after the APM or the HPM starts-up in the simulation mode. If automatic checkpointing is enabled, the simulation database is automatically saved at user-specified intervals. In addition to the standard database parameters, checkpointing a database in the simulation mode also saves the PVRAW and output parameters.



ATTENTION

When a checkpoint save occurs to the History Module, the existing checkpoint for that APM or the HPM node address is overwritten, even if the APM or the HPM is operating in a simulation mode.

If you want to save various stages of a simulation, perform the following:

- Checkpoint to removable media for each stage of the simulation. or,
- Checkpoint to the History Module and then copy the checkpoint files to user directories.

3.9 Effects of Checkpoint Save and Restore

Checkpoint save from actual IOP

When an actual IOP database (that is, a physical IOP database) is checkpointed, the checkpoint contains only "static" type data. Examples of the static data include user-entered values, such as ranges, alarm limits, setpoint limits, and filter values.

Checkpoint save from simulated IOP

When a simulated IOP database is saved, this static checkpoint database has the current PVRAW, and output values added to the checkpoint. These values are saved to enable capturing your current simulation state.

Checkpoint restore to actual IOP

When a checkpointed database is restored to actual IOPs, only the static data is restored. Actual IOPs do not need the checkpointed PVRAW data. Output checkpoint values are ignored. This means that an output returns to the previous IOP output value before a checkpoint restore. Mode values return to the default value of manual.

Checkpoint restore to simulated IOP

When a checkpointed database is restored into a simulation mode APM or HPM, the PVRAW, and output, if contained in checkpoint, is restored along with the static database.

Summary

To summarize the previous considerations, ensure that a checkpoint saved from an actual IOP and then restored to simulated IOP does not include the dynamic data. This means that you would have to set the dynamic values in simulated IOPs to the required values after you have restored the database with a checkpoint from actual IOPs.

CL/APM programs are not checkpointed

Note that the CL/APM programs are not part of the APM checkpoint. The CL/APM programs reside on HM volume &E##, where ## is the UCN network number.

3. Installation and Operation

3.9. Effects of Checkpoint Save and Restore

CL/HPM programs are checkpointed

Note that the CL/HPM programs are part of the HPM checkpoint. The CL/HPM simulation programs requiring modification must be recompiled and recopied to the appropriate directory. The CL/HPM programs reside on &E##, where ## is the UCN network number. The modified programs can then be checkpointed after they have been loaded to the HPM.

Checkpointing Summary

The overall relationship between database handling for checkpoint save/restore operations for the actual IOPs and the simulated IOPs is summarized in Table 11.

Table 11 Effects of Save and Restore

If the database is saved from...	Then the IOP database is saved as follows:	If this database is restored to...	Then the IOP database is restored as follows:
Actual IOP Database	• Static IOP database is saved • PVRAW is not saved • OP is not saved • Mode is not saved	Actual IOP database	• Static IOP database is restored • PVRAW = current field input • OP = previous value • Mode defaults to manual
Actual IOP database	• Static IOP database is saved • PVRAW is not saved • OP is not saved • Mode is not saved	Simulated IOP database	• Static IOP database is restored • PVRAW = NAN (Not a number, ---) • OP = -6.9% for analog outputs; • Off for digital outputs • Mode defaults to manual

3. Installation and Operation

3.9. Effects of Checkpoint Save and Restore

If the database is saved from...	Then the IOP database is saved as follows:	If this database is restored to...	Then the IOP database is restored as follows:
Simulated IOP database	• Static IOP database is saved • PVRW is saved • OP is saved • Mode is not saved	Actual IOP database	• Static IOP database is restored • PVRW = current field input • OP = previous value • Mode defaults to manual
Simulated IOP database	• Static IOP database is saved • PVRW is saved • OP is saved • Mode is not saved	Simulated IOP database	• Static IOP database is restored • PVRW = checkpointed value • OP = checkpointed value • Mode defaults to manual

3.10 Restoring the Database

Restoring simulation database to an on-process APM/HPM

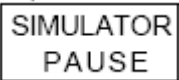
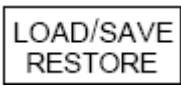
To restore a simulation database to the target APM or the HPM, which is controlling a process, the target APM or the HPM must first be idled. The simulation database can then be directly loaded following standard checkpoint restore procedures.

Restore Procedure

Prior knowledge of restoring a checkpoint to an APM or HPM is required. For additional information, see *Process Operations Manual*, SW11-601.

The Table 12 shows one of the ways to restore an APM or HPM's database. The display results are shown in Figure 26.

Table 12 Restore APM HPM Database

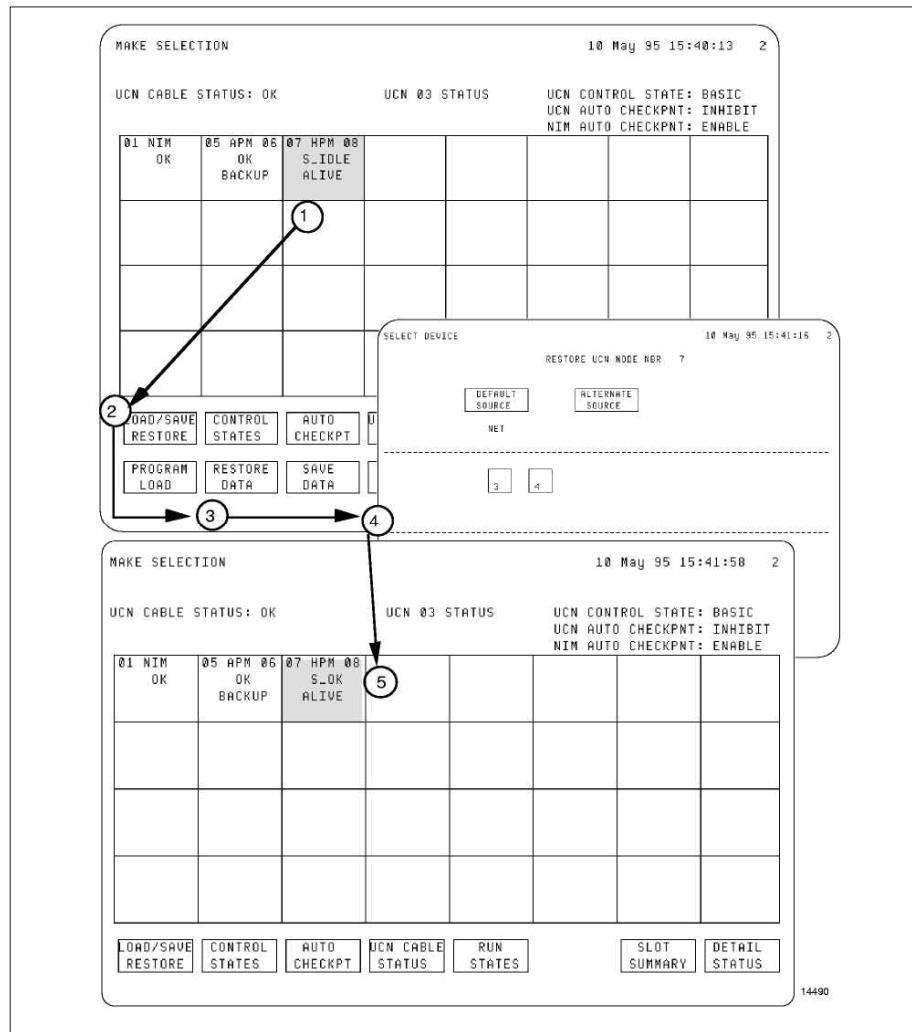
Step	Action
1	From the UCN Status Display, select the target representing the Primary APM or the Primary HPM that you want to load. Select either the  or the  target to place the APM or HPM in the proper state).  ATTENTION: The APM or HPM must be in the Idle (S_IDLE or S_PFIDLE) or Pause (S_PAUSE or S_PFPAUS) state to restore a checkpoint.
2	Select the  target.
3	Select the  target and press [ENTER].
4	Select the  target if your APM or HPM checkpoint is on a removable media.

Step	Action
5	Select the  target and press [ENTER].
6	After issuing a startup (or resume) command to the primary APM or the primary HPM, the status is displayed as S_OK/ALIVE. ATTENTION • If state is not S_OK, see Table 4 and Table 5 for status explanation. • When there is a hard failure of Primary HPM, the Secondary HPM takes over as Primary, and goes to the Idle state. Since the Secondary HPM does not have a database to run on, it displays the DB Invalid warning when trying to bring up the Secondary HPM to RUN State. Validate the database and then change the state to RUN state.

3. Installation and Operation

3.10. Restoring the Database

Figure 26 Example display results of Restore procedure



3.11 Saving the Database

Saving Checkpoints

If there are several starting points from which simulation could be resumed, several different checkpoints for the same APM or the HPM at different times must be saved. When saved to the History Module, each checkpoint overwrites the last copy. However, multiple checkpoints of the same APM or the HPM node can be saved by specifying an alternate destination(s) for storing each checkpoint on a removable media.

Save procedure

Prior knowledge of saving a checkpoint to an APM or an HPM is necessary. For additional information, see *Process Operations Manual*, SW11-601.

The steps in Table 13 show you one way to save an APM or an HPM's database. The displays you would use are shown in Figure 26, except that you would select the

SAVE
DATA

target for a checkpoint save.

Table 13 Save APM/HPM database

Step	Action
1	From the UCN Status Display, select the target representing the primary APM or HPM you want to save.
2	Select the LOAD/SAVE RESTORE target.
3	Select the SAVE DATA target and press [ENTER].
4	Select the ALTERNATE SOURCE target if the APM or the HPM checkpoint is on a removable media. Select the EXECUTE COMMAND target.
5	The status S_OK/ALIVE is to be expected with a primary APMM, or the APM, or the HPMM in the simulation mode.

3.12 Simulating the process

Simulating I/O

The basic function of the I/O simulator is to support another device which can act like the process to "stimulate" the I/O—that is, to read the outputs and to write to the inputs. This capability is enabled by the I/O simulator changing the access level for PVRAW stores to Operator. Based on this change, a user's I/O simulation program (in a VAX, Application Module, or even within the APM or HPM) can stimulate the input by writing directly to PVRAW. The PVRAW parameter can also be changed manually by an operator at a Universal Station (Figure 27).

Manual PV change

To manually change PVRAW, select the PVRAW parameter on an IOP Detail Display at the Universal Station and enter a new value. Note that the PVSOURCE remains in AUTO. Internally, the APM or the HPM transfers the manually entered PVRAW engineering unit value to PVCALC, then to the PV parameter that an operator can see.

While in Idle, the PVRAW is always displayed as "----" which means "Not a Number (NaN)." The user can set the PVRAW to a desired value during the "Idle" state. However, PVRAW is seen as NaN even after setting PVRAW to a value other than NaN. The desired value set by the user is shown correctly after the IOP state is set to run (OK).

Figure 27 Example of Changing the PV in Simulation Mode

The screenshot shows the IOP Detail Display for '01 M1 MIXER 1'. The display is divided into several sections: 'PVSOURCE selection' (with a box around 'PVSOURCE' and 'AUTO'), 'PVCALC' (60.0000), 'LASTPV' (60.0000), 'PVRAW' (60.5), and 'PVRAW' (60.0000). A box labeled '1' points to the 'PVRAW' parameter in the 'PVSOURCE' section. A box labeled '2' points to the 'PVRAW' value in the 'PVRAW' section. The display also shows 'RANGE LIMITS' and 'PV COEFFICIENTS'.

10 May 95 15:47:01 2

FI3000 PVSOURCE selection 01 M1 MIXER 1 FIRST PAGE

INDICATOR 0 60.0000 ALARM LIMITS POINT DATA

PVRAW 60.5 PVSOURCE AUTO PVHHTP 85.0 PTEXECST ACTIVE

60.0000 PVCALC 60.0000 PVHITP 70.0 ALENBST ENABLE

LASTPV 60.0000 PVLOT 20.0 OVERVAL 0

PVLLTP 10.0 1 ENTFORM FULL

PVROCTP 10.0 PVRAW 60.0000

PVROCNT 10.0 STSMG NONE

FI3000

GPM

FEED

SIMULATE 10.0

60.0 CONTCT OFF

PT TYPE ANINNIM

LCN NODE 24

PROC NET 03

UCN NODE 0

DEV TYPE H

MOD NUM 0

MOD TYPE H

SLOT NUM

ENTER

RANGE LIMITS PV COEFFICIENTS

PVEXEUHI 102.9 TF 0.00000

PVEUHI 100.0

PVEULO 0.0

PVEXEULO -2.9

PVRAWHT -----

1 After selecting the PVRAW parameter

2 Enter the simulated PV value into the port

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Programmed PV change

A variety of approaches are available in stimulating I/O. Some of the change approaches include:

- Process simulation program in a VAX
- Process simulation program in an Application Module
- Process simulation program in a CL/APM or CL/HPM program

CL/HPM statements – examples

The following example CL/HPM statements are valid, and show the parameters you can use in a process simulation (I/O stimulation) program.

Figure 28 Sample HPM program statements

```
SEQUENCE HPMSIM (HPM; POINT SIMLATR1)

EXTERNAL AI5, TIC22, DI21, DO21
PHASE START
STEP BEGIN

--TO PASS ENGINEERING UNITS TO ANALOG INPUT FROM
--AN OUTPUT, USE OP, CV, or OPEU PARAMETERS
--THIS EXAMPLE SETS ANALOG INPUT=REG. CONTROL OUTPUT
SET AI5.PVRAW = TIC22.OP * (TIC22.PVEUHI-TIC22.PVEULO)/100

--TO PASS BOOLEAN TO DIGITAL INPUT PV, USE PVRAW
SET DI21.PVRAW = DO21.SO

END HPMSIM
```

APM prefetch limits

Previous R400 prefetch limits (12 prefetches per step) apply to APMs operating on R500 and later releases.

HPM prefetch limits

Prefetch limits (12 prefetches per step) do not apply to HPMs.

3. Installation and Operation

3.12. Simulating the process

Performance

Estimated performance for supporting read/write access by upper-level process simulation program in a VAX or AM is 100-200 parameters per second. This estimated performance is in addition to the standard product throughput capability for display, history, and peer-to-peer data.

Peer-to-peer communication

Peer-to-peer communication is permitted when an APM or HPM is in a simulation mode. However, when an on control peer node attempts to communicate peer-to-peer with a node in the simulation mode, it has no way of "knowing" that the device is in a simulation mode. For that reason, Honeywell suggests that you lockout a simulation mode HPM from communicating with on control peer nodes (Note: This function is not available in the APM). In the HPM schedule and configuration display, you can enable the write lockout function. Note that HPM write lockout only prevents writes; nodes can still read (pull) data.

3.13 Transporting Databases

Introduction

Transporting databases between an actual physical APM (or HPM) and an APM (or HPM) in the Simulation Mode is possible in the following circumstances:

- Target and simulation mode APM (or HPM) are the same node.
- APM (or HPMs) on separate LCNs.
- APM (or HPMs) on same LCN, separate UCNs.

Target and simulation on same node (APM or HPM)

As implied throughout this document, the same checkpoint is available for use when the target APM (or HPM) and the simulation APM (or HPM) are the same. The following discussion does not imply that an APM's checkpoint can be loaded to an HPM, or an HPM's checkpoint can be loaded to an APM.

APM or HPMs on separate LCNs

The NIM and the APM (or HPM) checkpoints from an off-process simulation system can be transported to or from a target system on a separate LCN if the following conditions are true:

- The UCN Network numbers (Process Network numbers) on the separate LCNs are the same.
- The NIMs on the separate LCNs have the same LCN node number.
- The NIMs on the separate LCNs have the same UCN node number.
- The APM (or HPMs) on the separate LCNs have the same UCN node number.

3. Installation and Operation

3.13. Transporting Databases

Database Consideration

When database building occurs with APMs (or HPMs) on separate LCNs and separate UCNs, the following database considerations apply:

- Ideally, one of the two separate system databases is considered by the user as the "master" database. The "master" database is then used for starting up the other system. For example, the on-process system database and checkpoints are used to startup the off-process simulation system.
- To ease management of checkpoints, NIM and APM (or HPM) database modifications are checkpointed from the system the user regards as the "master" database. Then, at the user's convenience, the checkpoints are transported from one system to the other system to keep the databases synchronized.
- If new NIM points were built, the synchronization means rebooting the NIM and restoring the APM (or HPM) on the other LCN system with what the user regards as the "master" checkpoint. This step keeps the internal NIM point references the same for LCN configuration tasks (such as schematics and groups).
- If no new NIM points were built but changes were made to APM (or HPM) point parameters, the synchronization means restoring the APM (or HPM) on the other LCN system with the user's new "master" checkpoint.
- If parallel LCN database development (that is exclusive of NIM point building and NCF configuration) is occurring on both the off-process and on-process systems, the user takes the necessary steps (for example, Area Database Intermediate Data Files (IDF), recompiling pictures, Exception Build files, and so on) to synchronize the databases.

APM or HPM on same LCN, separate UCNs

Exception Build Files (EB files) from a Simulation Mode APM or HPM can be transported to or from a target APM or HPM on the same LCN, separate UCNs. The Exception Build files would have to be modified to change at least the following parameters:

- The UCN process network number.
- The APM (or HPM's) UCN node number.

You can conveniently change these values by using the replace function from the Text Editor.

Honeywell

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Phoenix, AZ 85027 USA